

# CropState-Vision: A Pilot Study of Six-Stage Rice Growth Classification from Field Images

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**Abstract**—Reliable recognition of rice growth stages is a prerequisite for decision-support systems that adapt agricultural information to crop development. This paper presents CropState-Vision, a pilot image-classification study that groups rice development into six BBCH-derived macro-stages and evaluates whether these stages can be distinguished from field images. Four baselines are compared on a 552-image collection: majority-class prediction, a shallow convolutional neural network trained from scratch, fine-tuned ResNet-18, and fine-tuned EfficientNet-B0. The shared test split contains 83 images. Evaluation includes accuracy, macro-F1, Mean Absolute Stage Distance (MASD), Expected Calibration Error (ECE), Brier score, bootstrap confidence intervals, exploratory paired comparisons, and a three-seed ResNet-18 check. The three learned models substantially outperform the majority baseline. ResNet-18 obtains the highest point estimates on the shared split, with 0.807 accuracy, 0.745 macro-F1, and 0.277 MASD. However, the pilot does not establish a statistically reliable advantage of one learned architecture over another after multiple-comparison correction. The results support the feasibility of six-stage rice classification, while also exposing important limitations: severe class imbalance, only six reproductive-stage test images, shared rather than architecture-specific hyperparameters, missing field and capture-session metadata, possible acquisition leakage, and the absence of independent agronomic label review. The study therefore reports feasibility rather than deployment readiness. Confidence-aware retrieval remains a future application and is not evaluated in this paper.

**Index Terms**—rice growth stage, BBCH, image classification, transfer learning, calibration, agricultural computer vision, pilot study

## I. INTRODUCTION

Agricultural management depends on crop development. Water management, nutrient application, disease surveillance, and harvest preparation differ across establishment, tillering, reproductive development, grain filling, and ripening. A digital decision-support system must therefore estimate the current crop stage before it can safely adapt downstream information.

Field images provide a practical input because phenological development is partly visible through leaf development, tiller formation, stem elongation, panicle emergence, grain formation, and ripening. Nevertheless, stage recognition

remains difficult near stage boundaries and under variation in viewpoint, lighting, plant density, field background, and camera distance. A model may also produce a confident but incorrect prediction, especially when a class is under-represented.

The broader CROPSTATE concept represents crop development as a maintained stage belief rather than a single static label. In that larger system, image classification is intended to provide visual evidence for later decision support. The present paper deliberately narrows the scientific scope to the upstream vision problem because the downstream retrieval experiments require agronomic relevance judgments from domain reviewers that are not currently available. This paper therefore asks whether the available image collection provides credible preliminary evidence that six rice macro-stages can be recognized.

The contributions are:

- 1) a six-class rice growth-stage formulation anchored to BBCH-derived macro-stages;
- 2) a controlled comparison of majority, shallow-CNN, ResNet-18, and EfficientNet-B0 baselines on one shared pilot split;
- 3) an ordinal and uncertainty-aware evaluation using macro-F1, MASD, ECE, Brier score, bootstrap intervals, paired analysis, and multi-seed checks;
- 4) a transparent analysis of data limitations that must be resolved before field-level or downstream decision-support claims are made.

## II. RELATED WORK

### A. Phenological Growth Stages and BBCH

The BBCH scale provides standardized numerical descriptions of observable plant development [1], [2]. Its first digit denotes a principal stage and its second digit denotes a more detailed secondary stage. In this study, fine-grained rice codes are grouped into six macro-stages for image classification. These groups are an experimental abstraction and do not replace the BBCH standard.

### B. Image-Based Rice Growth-Stage Recognition

Prior research has demonstrated that rice development can be estimated from field or aerial imagery using machine-learning methods [3]. However, reported performance

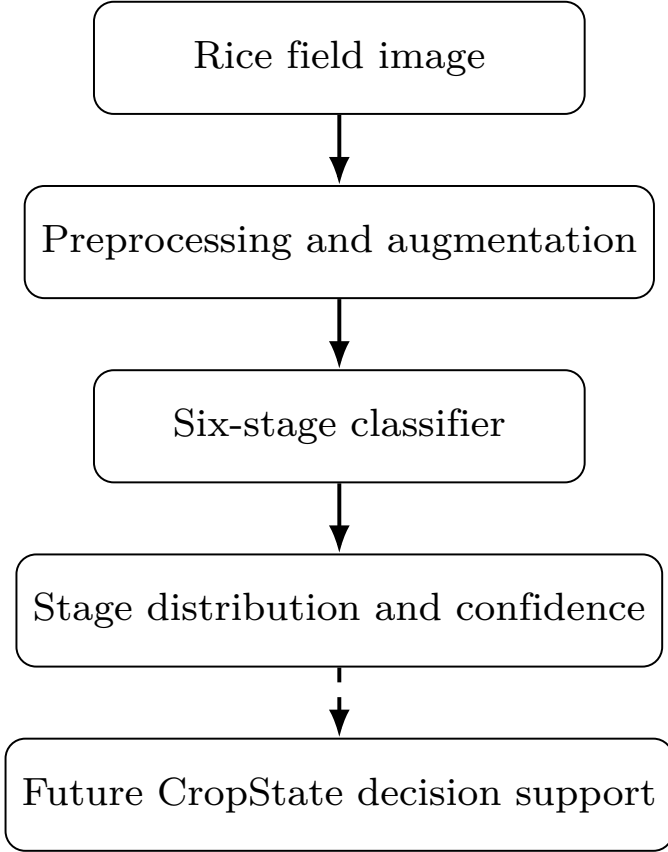


Fig. 1. Scope of the present study. The solid path is evaluated; downstream decision support is future work.

depends strongly on stage definitions, imaging conditions, sample size, and train-test separation. A visually accurate model on a random image split may still fail on a new field or season if it has learned background or acquisition cues.

### C. Transfer Learning and Confidence

Deep residual networks and EfficientNet provide widely used pretrained image-classification backbones [4], [5]. Transfer learning is particularly relevant when agricultural datasets are small. In addition to class accuracy, confidence quality matters because a downstream system may react differently to uncertain predictions. Temperature scaling and calibration metrics help assess whether softmax confidence is aligned with empirical correctness [6].

## III. TASK FORMULATION

### A. Six Rice Macro-Stages

Let  $\Phi = \langle \phi_1, \dots, \phi_6 \rangle$  be the ordered stage set in Table I.

For input image  $I$ , a model  $f_\theta$  produces logits  $\mathbf{o} = f_\theta(I)$ . The stage distribution is

$$p(\phi_i | I) = \text{softmax}(\mathbf{o}/T)_i, \quad (1)$$

TABLE I  
RICE MACRO-STAGES USED IN THE PILOT STUDY.

| Group                            | BBCH  | Label         |
|----------------------------------|-------|---------------|
| Germination and leaf development | 00–19 | Establishment |
| Tillering                        | 20–29 | Tillering     |
| Stem elongation and booting      | 30–49 | Stem/booting  |
| Heading and flowering            | 50–69 | Reproductive  |
| Grain development                | 70–79 | Grain filling |
| Ripening                         | 80–89 | Ripening      |

where  $T > 0$  is a temperature parameter estimated from validation data when calibration is applied. The predicted class is

$$\hat{\phi} = \arg \max_i p(\phi_i | I). \quad (2)$$

The full distribution is retained because adjacent stages may both be plausible and because later CropState modules may use uncertainty rather than only the top-1 label.

### B. Research Questions and Hypotheses

## IV. MATERIALS AND METHODS

### A. Pilot Dataset

The pilot collection contains 552 images divided into 386 training, 83 validation, and 83 test images for seed 42. Class counts in the training set are 24 establishment, 104 tillering, 38 stem/booting, 29 reproductive, 87 grain filling, and 104 ripening images. The reproductive class is particularly limited, with only six images in the test split.

Labels were inherited from stage-folder assignments. The collection has not yet received independent double annotation or formal agronomic adjudication. Field, capture-session, season, and reliable parent-image identifiers are unavailable in the current manifest. Consequently, the present split is effectively image-level and cannot guarantee that overlapping patches, near-duplicates, or images from the same acquisition context are isolated within one partition. The current results must therefore be interpreted as a feasibility estimate rather than a field-held-out benchmark.

A final dataset should record image identifier, parent-image identifier, field, capture session, date, season, variety, days after sowing, BBCH code, macro-stage, source, license, and annotation status. A corpus-wide quality audit should also detect corrupted exports, near-black images, severe dynamic-range compression, and duplicates before training.

### B. Compared Models

Four baselines are evaluated:

- **Majority class:** always predicts tillering, the largest test class;
- **Small CNN:** a shallow convolutional network trained from scratch;
- **ResNet-18:** an ImageNet-pretrained backbone fine-tuned on the pilot data;
- **EfficientNet-B0:** an ImageNet-pretrained lightweight backbone fine-tuned on the same split.

TABLE II  
RESEARCH QUESTIONS AND HYPOTHESES FOR THE SINGLE PILOT EXPERIMENT.

| ID  | Research question                                                                                                              | Hypothesis                                                                                                              |
|-----|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| RQ1 | To what extent can learned image classifiers distinguish six BBCH-derived rice macro-stages on the available pilot split?      | H1: Learned classifiers outperform the majority-class baseline in accuracy, macro-F1, and MASD.                         |
| RQ2 | Under one shared training recipe, is a fine-tuned pretrained backbone reliably better than a shallow CNN trained from scratch? | H2: Pretrained backbones achieve higher accuracy and macro-F1 than the shallow CNN.                                     |
| RQ3 | What error patterns and data limitations constrain interpretation of the pilot results?                                        | H3: Most ResNet-18 errors occur between adjacent stages, while rare stages exhibit lower recall and higher uncertainty. |

The three learned models use one shared training recipe, including the same epoch budget and freeze-then-unfreeze strategy. This provides a controlled engineering comparison, but it is not an architecture-at-optimum comparison because hyperparameters were not independently tuned for each model.

### C. Evaluation Metrics

Accuracy measures overall correctness, while macro-F1 gives equal weight to each class. Because the stages are ordered, Mean Absolute Stage Distance is

$$\text{MASD} = \frac{1}{N} \sum_{j=1}^N \left| \text{index}(\hat{\phi}_j) - \text{index}(\phi_j^*) \right|. \quad (3)$$

MASD distinguishes an adjacent-stage error from a prediction several stages away.

Confidence quality is summarized using Expected Calibration Error and the multi-class Brier score. Non-parametric bootstrap confidence intervals are computed from 10,000 resamples of the 83 test images. Exploratory paired comparisons use the stored per-image correctness vectors with Holm correction across six model pairs. Since correctness is binary, a confirmatory replication should additionally report McNemar’s test.

### D. Multi-Seed Check

ResNet-18 is also trained with seeds 7 and 123. In the present manifest, the seed affects both model initialization and image-level splitting. Therefore, the multi-seed table measures combined model and split variability rather than pure initialization sensitivity.

## V. RESULTS

All three learned models substantially exceed the majority baseline. ResNet-18 has the strongest point estimates, with 0.807 accuracy, 0.745 macro-F1, and 0.277 MASD. Small CNN follows with 0.759 accuracy and 0.702 macro-F1. EfficientNet-B0 reaches 0.711 accuracy and 0.614 macro-F1 under the same shared training recipe.

The learned models differ significantly from the majority baseline in the exploratory paired analysis. However, none of the three learned-model comparisons remains significant after Holm correction. Therefore, the pilot supports H1 but does not support the stronger H2 claim that a pretrained

architecture is reliably superior to the shallow CNN at this sample size.

ResNet-18 misclassifies 16 of the 83 test images. Twelve of these errors (75%) are adjacent-stage predictions, while four cross two or more stage boundaries. Its reproductive-stage recall is 0.167, compared with approximately 0.68–1.00 for the other five classes, but the reproductive class contains only six test images. EfficientNet-B0 has zero reproductive recall on the same six images. These values are better interpreted as evidence of class scarcity than as stable stage-specific effects.

The raw ResNet-18 softmax output has ECE 0.250, indicating that confidence is not yet sufficiently calibrated for high-stakes routing. EfficientNet-B0 has lower ECE but weaker classification performance, illustrating that calibration and discrimination are separate properties.

Accuracy ranges from 0.795 to 0.807 across three seeds. This is a mildly favorable stability signal, but it cannot separate model-initialization variance from split variance because the dataset lacks stable field or session groups.

## VI. DISCUSSION

### A. Answer to RQ1

The results support the feasibility of learning six rice macro-stages from the pilot images. All trained models outperform the majority baseline by a large margin, and their MASD values show that errors are generally closer to the true stage than majority prediction. Nevertheless, the result applies only to the present image-level split and cannot be interpreted as generalization to unseen fields or seasons.

### B. Answer to RQ2

ResNet-18 has the best point estimates, but the paired comparisons do not establish a reliable advantage over Small CNN after multiple-comparison correction. H2 is therefore not supported. The comparison also uses a shared recipe rather than independent hyperparameter optimization, which may particularly disadvantage EfficientNet-B0. The scientifically defensible conclusion is that all three learned models are viable pilot baselines, not that one architecture is definitively best.

TABLE III  
PILOT RESULTS ON ONE SHARED 83-IMAGE TEST SPLIT (552 IMAGES TOTAL; SPLIT SEED 42).

| Model                        | Accuracy | Macro-F1 | MASD ↓ | ECE ↓ | Brier ↓ | 95% CI (accuracy) |
|------------------------------|----------|----------|--------|-------|---------|-------------------|
| Majority class (tillering)   | 0.277    | 0.072    | 2.048  | n/a   | n/a     | [0.181, 0.373]    |
| Small CNN (from scratch)     | 0.759    | 0.702    | 0.434  | 0.189 | 0.400   | [0.663, 0.843]    |
| ResNet-18 (pretrained)       | 0.807    | 0.745    | 0.277  | 0.250 | 0.369   | [0.723, 0.892]    |
| EfficientNet-B0 (pretrained) | 0.711    | 0.614    | 0.422  | 0.183 | 0.421   | [0.614, 0.807]    |

TABLE IV  
EXPLORATORY PAIRED COMPARISONS ON THE SHARED TEST SPLIT (WILCOXON SIGNED-RANK TEST WITH HOLM CORRECTION).

| Pair                          | Accuracy difference | 95% CI           | Holm-adjusted $p$ |
|-------------------------------|---------------------|------------------|-------------------|
| Majority vs. ResNet-18        | −0.530              | [−0.639, −0.422] | < 0.0001          |
| Majority vs. Small CNN        | −0.482              | [−0.614, −0.349] | < 0.0001          |
| Majority vs. EfficientNet-B0  | −0.434              | [−0.554, −0.313] | < 0.0001          |
| ResNet-18 vs. Small CNN       | +0.048              | [−0.048, 0.145]  | 0.692             |
| ResNet-18 vs. EfficientNet-B0 | +0.096              | [0.012, 0.193]   | 0.137             |
| Small CNN vs. EfficientNet-B0 | +0.048              | [−0.060, 0.157]  | 0.692             |

TABLE V  
RESNET-18 ACROSS THREE SEEDS; EACH SEED ALSO CHANGES THE IMAGE-LEVEL SPLIT.

| Seed | Accuracy | Macro-F1 | MASD ↓ |
|------|----------|----------|--------|
| 42   | 0.807    | 0.745    | 0.277  |
| 7    | 0.807    | 0.802    | 0.398  |
| 123  | 0.795    | 0.789    | 0.289  |

### C. Answer to RQ3

ResNet-18 errors are mostly ordinally adjacent, which is preferable to arbitrary cross-stage confusion and supports the use of MASD alongside accuracy. However, non-adjacent errors remain and would be consequential in a downstream decision-support system. The reproductive stage is the clearest weak point, but its sample count is too small for a stable conclusion. The dataset design, rather than the model architecture alone, is currently the main constraint.

### D. Implications for CropState

The full stage distribution can serve as visual evidence in a future CropState state estimator. However, this pilot evaluates only image classification. It does not demonstrate that stage-conditioned retrieval, agronomic recommendations, or confidence-aware soft gating improve decision quality. Those claims require domain-reviewed knowledge chunks and relevance judgments from agricultural experts. They are therefore placed outside the scope of the current paper.

## VII. LIMITATIONS AND FUTURE WORK

The study has six major limitations. First, field, season, capture-session, and reliable parent-image metadata are missing, so the image-level split may contain acquisition or overlapping-patch leakage. Second, folder-derived labels

have not been independently reviewed against morphological criteria. Third, the dataset is imbalanced and the reproductive test subset contains only six images. Fourth, the same hyperparameters were used across architectures. Fifth, the multi-seed evaluation changes the split together with initialization. Sixth, the corpus lacks a documented image-quality and duplicate audit.

The next dataset revision should recover or collect parent-image, field, session, and season identifiers; place all related patches and burst images into one split; independently review a substantial labeled subset; balance rare stages; perform architecture-specific validation tuning; apply temperature scaling using validation logits; and evaluate once on a locked field- or season-held-out test set. Only after those requirements are met should the vision output be connected to expert-reviewed stage-aware retrieval.

## VIII. ETHICAL CONSIDERATIONS

Images collected from farms may reveal location, farm layout, or other sensitive context. Dataset releases should remove personal identifiers and precise geographic information unless explicit consent permits disclosure. The model is intended as a research component, not an autonomous agronomic authority. Low-confidence predictions should trigger uncertainty notices or requests for better images rather than definitive recommendations.

## IX. CONCLUSION

This paper reports a pilot evaluation of six-stage rice growth classification for the CropState research direction. On one 83-image test split, Small CNN, ResNet-18, and EfficientNet-B0 all outperform the majority baseline. ResNet-18 reaches the highest point estimates, but the current sample does not establish a statistically reliable superiority over the other learned models. Most ResNet-18 mistakes are adjacent-stage errors, while the rare reproductive class remains difficult. The results demonstrate

preliminary feasibility, not field-level readiness. Missing grouping metadata, possible leakage, class imbalance, unreviewed labels, shared hyperparameters, and incomplete calibration must be resolved before stronger claims are made. Downstream stage-aware retrieval remains future work requiring agricultural expert involvement and is not claimed by this study.

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